From drawing to cut list: a small storage cabinet

Before fabrication, each part must agree with the drawing and cut list. Work through one defined cabinet, check the arithmetic, then apply the same construction to new dimensions.

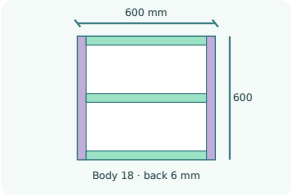
This lesson teaches fabrication planning. These dimensions are an original teaching example, not a universal standard or permission to operate machinery.

After this lesson you can

- Distinguish outside dimensions from individual panel dimensions.
- Calculate panels fitted between the sides, allowing for panel thickness.
- Calculate body depth when an overlay back is included in the overall depth.
- Prepare a numbered cut list and check quantities, units and assumptions.

Prerequisites: simple arithmetic and dimensions in millimetres. Tools: paper, pencil and a calculator, or the lesson calculator.

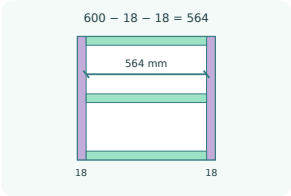
Read the brief first



An open-front cabinet with two full-height sides. The top, bottom and middle shelf fit between the sides. A separate back overlays the rear face. Overall depth includes the back. All dimensions are in millimetres.

Lesson assumptions: nominal thickness, butt-joint geometry, no grooves, doors, feet or finishing allowances. Any changed assumption requires a revised drawing and cut list.

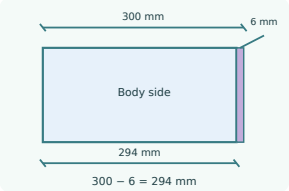
Calculate the internal width



Outside width is 600. Each side occupies 18 of that width. The top, bottom and shelf lengths are therefore 600 − 18 − 18 = 564. This equation applies because those panels sit between the sides in this construction.

$$W - 2t = 600 - 2 \times 18 = 564 \text{ mm}$$

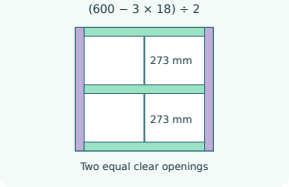
Separate body depth from back thickness



Overall depth is 300, including a 6-thick overlay back. Body panels are 300 − 6 = 294 deep. The back retains the full outside width and height: 600 × 600. A rebated or grooved back would require different geometry.

D − b = 300 − 6 = 294 mm

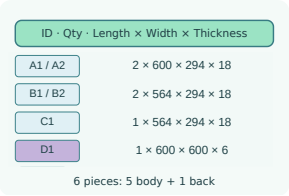
Check shelf and opening heights



The top, bottom and shelf occupy three thicknesses: 3 × 18 = 54. Remaining clear height is 600 − 54 = 546. Two equal openings are 546 ÷ 2 = 273 high each. This is the clear opening height, not the shelf length.

(H − 3t) ÷ 2 = 273 mm

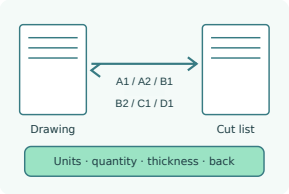
Turn the drawing into a cut list



Record part IDs, quantity, length, width and thickness. Group identical parts in a single row while retaining their IDs. Keep the thinner back as its own item. There are six pieces: five body panels and one back.

Net panel area alone does not determine stock-board quantity. A cutting layout must also account for material direction, saw kerf and allowances.

Review before releasing the list



Match every part ID to the drawing. Check units, quantities, thickness and back construction. Before fabrication, a competent specialist must approve cutting and finishing allowances, material direction, joints, fixing and intended loading. This lesson ends at preparing and checking the planning list.

Watching a video or passing the quiz does not establish competence to operate woodworking machinery; practical instruction, supervision and assessment are separate.

Knowledge check

How long is the top fitted between the sides?

- 600 mm
- 564 mm
- 582 mm

Subtract both side thicknesses: 600 − 2 × 18 = 564.

What is body-panel depth when the overall depth includes the back?

- 300 mm
- 306 mm
- 294 mm

The overlay back is included in overall depth: 300 − 6 = 294.

How many pieces are in this example?

- 6
- 5
- 7

Two sides + top + bottom + shelf + back = six pieces.

What is required before actual cutting?

- Approve only the generated picture
- Review drawings, allowances, material and fixings with a specialist
- Buy stock from net area alone

The teaching list contains no approved machining allowances, board layout, fixing or load specification.

Practice & assessment

New task: a cabinet 800 wide × 700 high × 350 deep. Body thickness is 18 and the overlay back is 6. Retain the same construction and two equal openings.

The automatic check below assesses arithmetic only. Drawings, part lists and documented assumptions need human review. This pilot does not award a certificate or fabrication competence.

Internal width

Body depth

Clear opening height

Review criteria: consistent drawing/list part IDs; correct dimensions and quantities; declared back construction and units; documented workshop approvals.

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Part	IDs	Qty	Length	Width	Body thickness
Sides	A1 / A2	2	600	294	18
Top & bottom	B1 / B2	2	564	294	18
Middle shelf	C1	1	564	294	18
Overlay back	D1	1	600	600	6

Net panel area alone does not determine stock-board quantity. A cutting layout must also account for material direction, saw kerf and allowances.

Review criteria: consistent drawing/list part IDs; correct dimensions and quantities; declared back construction and units; documented workshop approvals.